

## WORKPACK 6

# Spot Simple Patterns

Level 3 Data Technician · GreenWheels Urban Mobility · Self-Diagnostic Pack

## Get Ready to Smash This

A head-start check — this is not a test

Below are six things this pack covers. Tick the ones you can already do.

If you can tick all six: jump to the Mini-Mission on page 7 and prove it.

- ☐ I can sort a dataset to find the highest and lowest values in any column.
- ☐ I can read which station ranks first and last after sorting by total hires.
- ☐ I can compare zone totals by grouping records and reading the numbers.
- ☐ I can spot a seasonal pattern by comparing Q1 and Q3 values in the same column.
- ☐ I can write a plain-English sentence that describes one pattern from the data.
- ☐ I know the difference between an observation (what I can see) and an interpretation (what I think it means).

**Priya at the review meeting says:** "Marcus sent me a table with 40 rows and said: Here is the analysis. I don't know what to do with this. Give me one sentence about what it shows." This pack teaches you how to give her that sentence.

## What This Pack Is About

Packs 1 to 5 were about getting data clean and correctly shaped. This pack is about reading what the clean data actually shows. It does not use formulas or pivot tables. It uses sorting, grouping, and your eyes.

The goal is simple: look at the numbers, spot something true, and write it down in one clear sentence. This is the first step in Analytical Direction — the second part of the Scenario Demonstration.

## Observations and Interpretations

An observation is something you can see directly in the data. 'Central Square has the highest total hires of any single station' is an observation — you can verify it by sorting.

An interpretation is what you think the observation means. 'Central Square performs best because it is near the city centre' is an interpretation — it uses context, not just the numbers. This pack focuses on observations. Interpretations come later, in the oral part of the demonstration.

### NOTE

**The pattern sentence formula:** I notice [what] because [the data shows]. For example: 'I notice that total hires are highest in Q3 across every zone, because sorting by quarter and summing hires shows Q3 as the largest figure in four out of five zones.'

## The Skills

Open Pack6\_GW\_data.xlsx. All five exercises use this one file. You do not need Power Query for this pack — work directly in the Excel spreadsheet. You will sort, filter, and read numbers.

**1**

### Sort to Rank

*Which station performs best and worst?*

#### SCENARIO

*"I need to know which of our 10 stations had the most hires across the whole year," says Priya. "And which had the least. I don't need a chart — just sort and tell me."*

Sorting a column puts rows in order from smallest to largest (ascending) or largest to smallest (descending). When you sort by Total Hires descending, the station with the most annual hires floats to the top. Reading the first few rows and the last few rows shows the range.

#### WORD TO KNOW Annual total

The sum of all quarterly records for one station across the year. The dataset has four records per station (Q1 to Q4), so sorting by Total Hires shows the best individual quarters — not the best station overall. To see station totals, you sum the four rows per station.

There are two ways to see station totals without a pivot table: (a) sort by Station Name then by Total Hires, and add up the four rows in your head; or (b) add a SUMIF formula in a spare column: =SUMIF(B:B, B2, F:F) will sum all Total Hires for the station in B2.

#### TRY IT

Sort the dataset by Total Hires descending. Which single record has the highest quarterly hires? Which station has it, and which quarter?

Now sort by Station Name ascending, then by Total Hires descending (multi-level sort: Data → Sort, add a level). Look at the four rows for Central Square. What are the four quarterly hire counts? What is their rough total?

Looking at the station totals by eye: which station appears to have the highest annual total hires? Which appears to have the lowest? Write one sentence for each.

#### CHECK YOURSELF

Central Square's four quarterly hire counts are 512, 647, 689, and 495. Without adding them up exactly, which quarter was highest, and what does that suggest about when demand peaks?

**Answer:** Q3 (689) was highest, then Q2 (647), Q4 (495), Q1 (512). The summer quarter has the most hires. This suggests demand at Central Square peaks in summer. All four quarters are above 400 hires, so this is consistently a high-performing station.

**2**

### MIN and MAX

**Finding the extremes without scrolling****SCENARIO**

*"I want to know the highest and lowest customer satisfaction scores in the dataset," says Brian. "And I want to know which station has the worst average duration. Can you just tell me the numbers?"*

MIN and MAX are the simplest analytical tools in Excel. MIN returns the smallest value in a range; MAX returns the largest. They are faster than sorting and tell you immediately where the extremes are.

**WORD TO KNOW**  
**MIN / MAX**

Simple Excel formulas: =MIN(range) returns the smallest value, =MAX(range) returns the largest. Use them to instantly find the extremes of any numeric column without sorting or scrolling.

**WORD TO KNOW**  
**Conditional formatting for extremes**

Home tab → Conditional Formatting → Top/Bottom Rules → Top 1 item (or Bottom 1 item). Highlights the extreme value in a colour so it is visible at a glance without sorting.

To use MIN/MAX: click a blank cell outside the data (for example, cell L2). Type =MAX(J2:J41) to find the highest Customer Satisfaction Score across all 40 records. Type =MIN(J2:J41) to find the lowest.

**TRY IT**

In a blank cell, type =MAX(J2:J41) to find the highest satisfaction score. Then type =MIN(J2:J41) for the lowest. What are the two values?

Use Conditional Formatting → Top/Bottom Rules → Top 1 Items on the Customer Satisfaction Score column. Which station and quarter is highlighted?

Now find the minimum Avg Hire Duration (Mins) across all records using =MIN(G2:G41). Which record has the lowest duration? Is that the same record with the lowest satisfaction score?

**CHECK YOURSELF**

Customer Satisfaction Score runs from 3.5 to 4.5 in this dataset. What observation can you make about the range?

**Answer:** The range is only 1.0 — scores are tightly clustered between 3.5 and 4.5. No station scored below 3.5 or above 4.5. This suggests satisfaction is consistently moderate-to-good across all stations, with no extreme outliers in either direction.

**3****Compare Zones***Which zone brings in the most hires?***SCENARIO**

*"I want a rough league table — not a full pivot — just: which zone is busiest, which is quietest," says Priya.*

*"Use the data as it is. No pivot. Just filter and add."*

Without a pivot table, you can compare zones by filtering for one zone at a time and reading the values, or by sorting by Zone and doing rough mental maths on the grouped rows.

**WORD TO KNOW****AutoFilter for one zone**

Click the filter arrow on the Zone column → tick one zone only → click OK. Only records for that zone are visible. You can then read the Total Hires for those records. Clear the filter to return to the full data.

The five zones in the dataset: Central (4 stations, 16 records), North (2 stations, 8 records), East (2 stations, 8 records), South (1 station, 4 records), West (2 stations, 8 records). Zone totals are not directly comparable by row count alone — Central has more stations, so naturally more rows.

**NOTE**

For a fair zone comparison, the right metric is average hires per station-quarter (total hires for the zone, divided by the number of records). This accounts for the fact that some zones have more stations than others.

**TRY IT**

Filter the Zone column for 'Central'. Write down the Total Hires values for all 16 rows. Do not add them up exactly — just look at whether they are mostly high, medium, or low. Clear the filter.

Repeat for 'North'. Compare the range of values in North with the range in Central. Write one sentence: which zone has larger individual hire counts?

Now filter for 'East'. Compare East records to North records. Are East hire counts higher or lower than North hire counts on average?

**CHECK YOURSELF**

Central zone has 4 stations and 16 records. North has 2 stations and 8 records. If Central's total hires are 8,392 and North's are 3,455, which zone has a higher average per station-quarter?

**Answer:** Central:  $8,392 \div 16 = 524.5$  per record. North:  $3,455 \div 8 = 431.9$  per record. Central has higher average hires per station-quarter, not just a higher total. The total is also higher, but that could simply be because it has twice as many stations.

**4****Spot a Seasonal Pattern***Is Q3 always the busiest quarter?***SCENARIO***"Every year we expect summer to be our busiest season," says Priya.**"But does the data actually show that? Can you check whether Q3 is the highest quarter at most stations, without building a chart?"*

A seasonal pattern means that values rise and fall in a predictable way that repeats with the time of year. To check for one in this dataset: sort or filter by Period and compare the hire counts. If Q3 hires are consistently higher than Q1 hires across most stations, that is a seasonal pattern.

**WORD TO KNOW****Seasonal pattern**

A trend that repeats based on the time of year. Hire schemes commonly peak in summer (Q3) due to weather. To check: compare the same metric (Total Hires) across different time periods (Q1, Q2, Q3, Q4) for the same stations.

To check Q3 vs Q1 for each station: sort by Station Name ascending, then by Period ascending. This groups all four quarters for each station together. Read the Q1 and Q3 rows side by side for each station.

**TRY IT**

Sort by Station Name ascending, then Period ascending (Data → Sort, two levels). Look at Central Square rows 1–4. Compare Q1 (512 hires) with Q3 (689 hires). Is Q3 higher? By how much?

Now check Victoria Park: Q1 = 318, Q3 = 445. Is Q3 higher? Check North Bridge: Q1 = 187, Q3 = 268. Is the pattern consistent?

Look across all ten stations. At how many stations is Q3 the highest-hire quarter? Write one sentence: 'I notice that Q3 is the highest-hire quarter at [n] out of 10 stations, which suggests...'

**CHECK YOURSELF**

At Central Square, hires go: Q1 = 512, Q2 = 647, Q3 = 689, Q4 = 495. What is the pattern and what observation does it suggest?

**Answer:** Hires rise from Q1 to Q3 then drop in Q4. Q3 is the peak. The observation: 'Hires at Central Square rise through the first three quarters, peak in Q3 (summer), then fall sharply in Q4, suggesting demand is higher in warmer months.'

**5****Write the Observation***Turning numbers into one clear sentence***SCENARIO**

*"I can see what the data shows," says Marcus. "I just don't know how to say it."*

*Priya replies: "Start with: I notice. Then say what you see. Then say because and point to the numbers. That's it."*

An observation sentence has three parts: what you noticed, what the data shows, and why the data supports what you are saying. You do not need fancy language. Plain English, specific numbers, and a clear claim.

| Quality      | Example                                                                                                                                       | Why it works                                 |
|--------------|-----------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------|
| Too vague    | The Central zone does well.                                                                                                                   | No number. Nothing to verify.                |
| Too detailed | Central Square in Q3 2024 had 689 hires which is 177 more than Q1 and represents a 34.6% increase.                                            | Too much. One sentence should give one idea. |
| Just right   | I notice that Q3 is the highest-hire quarter at 9 out of 10 stations, which suggests a consistent seasonal peak in summer across the network. | Specific, verifiable, one idea.              |

The three building blocks of a good observation sentence:

**NOTE**

- 1. What:** Name the pattern or fact you noticed.
- 2. Evidence:** Include the number(s) from the data that prove it.
- 3. Signal:** Say what the pattern suggests — not what causes it, just what it points to.

**TRY IT**

Write an observation sentence about station performance (use what you found in Chunk 1). Use the formula: 'I notice that [station] has [value] which is [higher/lower] than [comparison], suggesting [what this shows].'

Write an observation sentence about seasonal patterns (use Chunk 4 findings). Mention at least one specific number.

Write an observation sentence about zone performance (use Chunk 3 findings). Compare two zones by name.

**CHECK YOURSELF**

What is wrong with this observation: 'The data shows that some stations are busy and some are not.'

**Answer:** It contains no specific numbers, no station names, no time period, and no comparison. It could describe any dataset. A good observation is verifiable — someone reading it should be able to check the claim against the data without asking you any questions.



## Spot the Mistake

Can you see what went wrong?

Marcus sorted Total Hires descending. The top row shows HC-015: Market Street, Q3 2024, 1,847 hires. He writes: 'I notice that Market Street is the busiest station, with 1,847 hires in Q3 2024 — higher than any other station or quarter in the dataset.'

Priya reads it and says: 'That is wrong. And I don't mean the observation is wrong — the data is wrong. Did you validate this dataset before you analysed it?'

### WATCH OUT

HC-015 is the anomaly flagged in Pack 2 — 1,847 hires is more than double the valid range of 100–800. An observation built on unvalidated data is worse than no observation: it is confidently wrong. Always validate before you analyse. This is why Packs 2 and 3 come before Pack 6.

### CHECK YOURSELF

Pack 6 uses Pack6\_GW\_data.xlsx, not the original messy file. Why is that important?

**Answer:** Pack6\_GW\_data.xlsx is the clean dataset — all eight anomalies corrected, duplicate and blank records removed. Analysing the messy file would produce wrong observations. Using the clean file ensures every pattern you spot is real, not the result of a data error.

## Mini-Mission

Two minutes — prove it

Open Pack6\_GW\_data.xlsx. Without referring back to the chunks, write five observation sentences:

1. One about the station with the highest total hires across the year.
2. One about the station with the lowest total hires across the year.
3. One about the seasonal pattern in Q3 versus Q1.
4. One about the highest customer satisfaction score.
5. One about the difference between any two zones.

Each sentence must contain at least one specific number from the data. Swap with someone and check each other's observations against the data.

## You Can Now Answer

These are the kinds of questions that come up in the oral part of the Scenario Demonstration

- ☐ Which station has the highest annual total hires, and how do you know?
- ☐ What is the highest single-quarter hire count, and what station does it belong to?
- ☐ Is there a seasonal pattern in hire demand? What evidence supports your answer?

- ☐ Which zone appears to have the highest average hires per record?
- ☐ What is the range of customer satisfaction scores, and what does that range suggest?
- ☐ What is the difference between an observation and an interpretation?

## If You Got Stuck

No problem — here is what to practise next

*The hardest part is not finding the pattern. It is writing it down simply enough that someone who has not seen the data can understand it. If your sentence needs more than twenty words, try again. Simpler is almost always clearer.*

— **Brian, GreenWheels Analyst**

If writing the observation sentence felt hard: practice with one number at a time. Find the highest satisfaction score. Write: 'The highest customer satisfaction score is [X], recorded at [station] in [period].' That is already a valid observation — it is specific and verifiable.

If spotting seasonal patterns was difficult: focus on just one station. Compare its Q1 and Q3 hire counts. If Q3 is higher, you have a seasonal observation for that station. Then check two more stations and see if the pattern holds.

Well done for completing all six packs. You have validated a dataset, cleaned it in Power Query, added calculated columns, merged two data sources, and practised reading patterns — all the core skills for Scenario Demonstration 2.